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THE UNITED REPUBLIC OF TANZANIA
IKANA SECONDARY SCHOOL
FORM THREE TERMINAL EXAMINATION
033/1 BIOLOGY 1

TIME: 3 Hours ***Friday, 30th May 2025 a.m.***

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.

2. Answer **all** questions in sections A and B and **two (2)** questions from section C.

3. Section **A** carries **sixteen (16)** marks, section **B** carries **fifty-four (54)** marks and section **C** carries **thirty (30)** marks.

4. All writing should be in **blue** or **black** pen, except for diagrams that must be drawn in pencil.

5. Communication devices and any unauthorised materials are **not** allowed in the examination room.

6. Write your **name** on every page of your answer answer sheet (s)

SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) - (x) choose the correct answer from among the given alternatives and write its letter beside the item number.

(i) The structure that prevents food from entering the wind pipe is

A Glottis
B Tongue
C Soft palate
D Epiglottis
E Mouth

(ii) Cell membrane, nucleus and cytoplasm make the plant and animal cell to be similar. Why do plant and animal cells possess cell membrane?

A To synthesize proteins
B To control all activities of the cell
C To provide energy to the cell
D To regulate movement of substances in and out of the cell
E To store genetic information

(iii) Which of the following types of vertebrate muscles would you find in the heart?

A Smooth muscle
B Cardiac muscle
C Striated muscle
D Skeletal muscle
E Voluntary muscle

(iv) An amoeba moves by

A producing pseudopodia
B using cilia
C using flagella
D the flow of ectoplasm
E producing limbs

(v) Which one of the following hormones controls reabsorption of water from the urinary tubules when the amount of water in the blood is below normal?

A Adrenaline
B Luteinizing hormone
C Oxytocin
D Anti-diuretic hormone
E Aldosterone

(vi) A neuron that carries sensory information towards the central nervous system is called

A Interneuron
B Efferent neuron
C Afferent neuron
D Motor neuron
E Relay neuron

(vii) The following is not true about viruses

A They attack living cells
B Multiply rapidly inside living cells
C They are easy to destroy
D They spread through the air
E They cause diseases

(viii) A term used in Biology to identify the organism that depends on host for food but not shelter is known as:

A Parasite
B Endoparasite
C Symbiosis
D Ectoparasite
E Mutualism

(ix) Which safety precaution should be taken when administering first aid to a wounded person?

A Washing hands with soap
B Wearing protective gloves
C Calling the ambulance for pick-up
D Washing the wound with soap
E Drying the wound with clean cloth

(x) Which of the following is correct for someone with a nose bleed?

A To move around with their head tilted up
B To sit with their head tilted to the side
C To sit quietly with their head in an upright position
D To sniff in between intervals and sit with their head tipped forward
E To sit with their head tipped forward

2. Match the responses in **List B** with the word/phrases in **List A** by writing the letter of the correct response beside the item number.

List A	List B
(i) The growth movement in response to gravity.	A Hydrotropism
(ii) The growth movement of a plant part in response to unilateral source of light.	B Thigmotropism
(iii) Movements shown by plants such as leaves, stems and roots in response to different stimuli.	C Geotropism
(iv) Induced movement controlled by external factors such light, water, chemicals, touch and gravity.	D Movement of curvature
(v) Plant movement which are determined by the direction of the stimulus.	E Autonomic movement
(vi) The growth movement of a part of a fixed plant in response to touch.	F Paratonic movement
	G Phototropism
	H Tropic movement

Section B (54 Marks)

Answer **all** questions in this section

3. (a) Give two reasons why biologists classify organisms.

(b) The following organisms belong to the Division angiospermophyta: cotton, sisal, maize, sugarcane, hibiscus, and bean plants. This division is further divided into two (2) classes.

(i) Group the organisms according to their respective classes.

(ii) Write down four (4) differences between the maize and bean plants in terms of leaf shape, seed structure, type of root system and arrangement of veins in the leaf.

4. (a) Briefly explain why do Biologists prefer to use a light microscope in the laboratory rather than using normal eyes?
(b) Draw a well labelled diagram of light microscope.

5. The diagram in figure **1** is a reflex arc in human being. Study the diagram and answer the questions that follow:

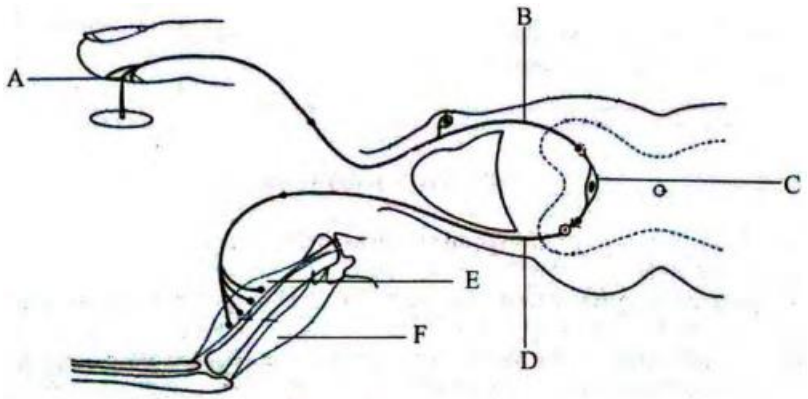


Figure 1

(a) Name the parts labelled A-F
(b) Outline the function of parts A, B, C and D.
(c) If some one touches a hot object, what is the response shown by part E?

6. In an adult education class, the teacher emphasizes the need for eating a balanced diet; but one of the participants did not understand. You are now asked to clarify this for him. Use the following headings in giving the clarification.
(a) Meaning of a balanced diet
(b) Advice the participant any four importance of a balanced diet and three effect of not eating a balanced diet

7. (a) Differentiate between excretion and egestion as used in Biology.
(b) Excretory products eliminated by plants are very useful to both plants and animals.
(i) Give any four types of excretory products eliminated by plants.
(ii) In what ways are plant excretory products mentioned in 7b (i) useful?

8. (a) Distinguish between exoskeleton and endoskeleton using the following subheadings:
(i) Position in the body
(ii) Composition
(iii) Examples of organisms in which they are found.
(b) In three points, explain the function performed by the human skeleton.

SECTION C (30 Marks)

Answer **two (2)** questions from this section.

9. The brain is the largest portion of the nervous system and very important for controlling activities in the body. With the aid of a diagram, describe the functions of the different parts of the human brain. Explain why only humans are said to be intelligent.

10. Describe the functions of the major components of blood.

11. With at least two examples, briefly explain six methods by which human infectious diseases are transmitted.